



by Kunal Sir

Quick Overview To The JavaScript.

❖ JavaScript: -

JavaScript is high level, just in time compiled programming language.

JavaScript is use for **client-side scripting**.

❖ Difference Between Server-Side Scripting & Client-Side Scripting.

Sr No	Server-Side Scripting	Client-Side Scripting
1	Server-side scripting is done at the backend Application, where the source code is hidden to the user.	Client-side scripting is done at the front end of Application, it is accessible to the user and it can be manipulated by users to.
2	Server-side scripting is done to hide the business logic from user	Client-side scripting is done to reduce the load on server.
3	Server-side scripting execute slower as compare to client-side scripting	Client-side scripting execute faster as compare to server-side scripting
4	Server-side scripting is secure as compare to client-side scripting as users can't access it directly	Client-side scripting is less secure as compare to sever-side scripting as users can access it directly

❖ Where To Write JavaScript: -



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1) In `<body>` tag: Inside the `<script>` tag we can write JavaScript code.

```
index.html X
day1 > index.html > html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Document</title>
7  </head>
8  <body>
9      <h1>Welcome to JavaScript</h1>
10     <script>
11         console.log("Java Script In Body..!")
12     </script>
13 </body>
14 </html>
```

2) In `<head>` tag: Inside the `<script>` tag we can write JavaScript code.

```
index.html X
day1 > index.html > html > head > script
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Document</title>
7      <script>
8          console.log("Java Script In Body..!")
9      </script>
10 </head>
11 <body>
12     <h1>Welcome to JavaScript</h1>
13 </body>
14 </html>
```



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3) External JS file: Here, we need to create a file with `.js` extension, and this file we can link to our html page using `<script>` tag.

Step 1:

The screenshot shows the VS Code Explorer on the left with a folder named 'day1' containing 'index.html' and a newly created 'main.js' file. The main editor area shows the 'main.js' file with a single line of code: `console.log("JavaScript in External file..!");`

Step 2:

The screenshot shows the VS Code Explorer with 'index.html' selected. The main editor area displays the HTML code for 'index.html', which includes a `<script src="main.js"></script>` tag in the head section, linking to the external JavaScript file.



❖ JavaScript Variables / Data members: -

Variables are containers for storing data.

Variables are nothing but the name given to the memory location where the actual is stored.

JavaScript variables can be declared in 4 ways.

1. Automatically
2. Using var keyword (1995)
3. Using let keyword (2015, ES6)
4. Using const keyword (2015, ES6)

■ **Automatically: -**

The variable declared **without using var, let or const keyword** are automatically declared variables. They are automatically **declared after their first use.**

Example:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-
width, initial-scale=1.0">
  <title>Document</title>
  <script>
    x=10;    // automatically declaration
    var x="sagar"; // re-declarartion
```



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```
        console.log(x); //output:- sagar
    </script>
</head>
<body>

</body>
</html>
```

Note: - It is considered as a good programming practice to always declare variable before use.

▪ Using `var` Keyword: -

The var keyword is used in all JavaScript code from 1995 to 2015. The var keyword should only use if the code is written for old browsers.

The variable defined using var keyword **can be re-assign**.

The variable defined using var keyword **can be redeclare**.

The variable defined using var keyword **have global scope**.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-
width, initial-scale=1.0">
    <title>Document</title>
    <script>
```



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```
var name="Pratik";    // automatically
declaration
var name="CJC"; // re-declarartion
console.log(name);// output:- CJC
function m1()
{ // gloabl scope variable can be
asscess in functions.
    console.log(name); //output:- "CJC"
}
m1();
name="cjc by kunal sir." // re-assign
console.log(name); // output:- cjc by
kunal sir.
</script>
</head>
<body>

</body>
</html>
```

▪ Using `let` keyword: -

The let keyword is added in JavaScript in 2015 as per ES6 guidelines.

Variable define with let keyword **can be re-assign.**

Variable define with let keyword **cannot be redeclare.**

Variable define with let keyword **have block scope.**

Example: -



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```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-
width, initial-scale=1.0">
  <title>Document</title>
  <script>
    let name="Pratik";    // automatically
declaration
    let name="CJC"; // redclartion is not
possible in case of let
    console.log(name);// output:- CJC
    function m1()
    { // example of block scope.
      let name="CJC,Pune"
      console.log(name); //output:- "CJC"
    }
    m1();

    console.log(name); // output:- CJC
  </script>
</head>
<body>

</body>
</html>
```



■ Using `const` keyword: -

The const keyword is introduced in JavaScript in 2015 as per ES6 guidelines.

Using const keyword we can achieve **constant** in JavaScript.

Variable define with const keyword **must be assigned**.

Variable define with const keyword **cannot be re-assign**.

Variable define with const keyword **cannot be redeclare**.

Variable define with const keyword **have block scope**.

Example: -

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-
width, initial-scale=1.0">
  <title>Document</title>
  <script>
    const name="Pratik";    // automatically
declaration
    // var name="CJC"; // redclartion is not
possible in case of let
    console.log(name);// output:- CJC
    function m1()
    { // example of block scope.
```



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```
const name="CJC,Pune"
console.log(name); //output:- "CJC"
}
m1();
// name="cjc by kunal sir."
console.log(name); // output:- cjc by
kunal sir.
</script>
</head>
<body>

</body>
</html>
```

❖ Difference between var, let and const: -

SR	Features	var keyword	let keyword	const keyword
1	Introduced on	1995	2015	2015
2	Re-Declaration	Yes	No	No
3	Must Assign value	No	No	Yes
4	Re-Assignment	Yes	Yes	No
5	Variable scope	Global	Block	Block